CSC 225 — Sorting Robot Lottery

Coding mini-assignment | 50 points

The sorting robots say they are overworked. Your job is to keep comparison-count records for their 50 shifts. The campus has requested fewer spreadsheets and fewer robot excuses.

1. Draw two sorting methods — 5 points

Write **choose_algorithms(seed)** using **random.Random(seed).sample(...)** to choose two distinct methods from Bubble, Selection, Merge, Quick, and Heap. Save the seed; no redraws.

2. Build a TimeSheet class — 20 points

Store the algorithm name and a list of comparison counts. Implement **__init__**, **add_run**, **average**, **median**, **minimum**, **maximum**, and **standard_deviation**. Use the **population** standard deviation (divide by N). Statistics return **None** for an empty list. Do not use the statistics module.

3. Implement the two selected sorts — 20 points

Only implement your assigned two functions. Sort the supplied list **in place** in ascending order and return **(values, comparison_count)**. Count comparisons of **data values**, not loop or index checks. No built-in sort or sorted inside the sorting functions.

4. Run 50 random lists — 5 points

Run **python3 Sorting_Robot_Starter.py**. The driver makes 50 random lists of 100 integers and gives the two algorithms copies of each same list. It records comparisons **without automated correctness checks**. Put all 50 counts for each algorithm and the five statistics in **Sorting_Robot_Results.md**. Identify the algorithm using fewer total comparisons. No Big-O or timing analysis.

5. Fork, clone, and publish your work on GitHub

Fork: your own GitHub copy of the instructor's starter repository, where you can edit without changing the original. **Clone:** download your fork onto your computer.

Step 1: Open the instructor-provided starter repository link on GitHub. Click **Fork**, select your account, choose **Create fork**, and keep it public.

Step 2: On **your fork**, click **Code** and copy the HTTPS URL. Open a terminal:

```
git clone https://github.com/YOUR-USERNAME/REPOSITORY-NAME.git
cd REPOSITORY-NAME
```

Replace the example URL and folder name with your actual fork. The instructor will provide the original repository link.

Step 3: Edit the two files in the cloned folder, then run **python3 Sorting_Robot_Starter.py** and record your output.

Step 4: Save your work to GitHub from the cloned folder:

```
git status
git add Sorting_Robot_Starter.py Sorting_Robot_Results.md
git commit -m "Complete Sorting Robot Lottery"
git push origin HEAD
```

Step 5: Open your fork on GitHub to verify the latest files and commit. **Submit the URL of your public fork** in the LMS. You do not need a pull request.

Reminder: Fork = your remote GitHub copy; clone = local copy; push = upload committed changes to your fork. No third-party Python packages are required.

Example statistic: [6, 2, 9, 3] has mean 5, median 4.5, min 2, max 9, and population standard deviation approximately 2.739.